INCH-POUND

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MILITARY SPECIFICATION

RELAY ASSEMBLY: GUN-FIRING, WATERPROOF, 24 VOLT DC

This specification is approved for use within the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of waterproof gun-firing relay assembly with a nominal rating of 24 volts (V) direct current (DC) for use in the main gun and machine gun circuits of military combat vehicles. The assembly consists of relays which control the firing of the main gun by the operation of a time delay circuit and a separate fast-operating, fast-release relay which controls the firing of the machine gun (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5945

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SPECIFICATIONS
FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-S-571	-	Solder; Tin Alloy; Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-M-14	-	Molding, Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-997	-	Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass, Cloth, Silicon Resin.
MIL-S-8660	-	Silicon Compound, NATO Code Number S-736.
MIL-P-15037	-	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.

STANDARDS
FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.
FED-STD-595	-	Colors.

MILITARY

MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-193	-	Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1184	-	Electrical Components for Automotive Vehicles; Waterproofness tests.
DOD-STD-1866	-	Soldering Process General (Non-Electrical).
MIL-STD-45662	-	Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

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2.1.2 Government drawing. The following Government drawing form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be the one in effect on the date of the solicitation.

DRAWING ARMY		
8338636	-	Relay Assembly, Gun Firing

(Copies of specifications, standards, and drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4).

3.2 Materials. Materials shall be as specified herein, on applicable standards or drawing, or in applicable specifications (see 4.8.1).

3.2.1 Plastic material. Plastic material shall conform to to MIL-M-14 for molded material, MIL-P-997 or MIL-P-15037 for laminated material.

3.2.2 Metals. Metals shall be of a corrosion-resistant type or shall be treated to resist corrosion.

3.2.2.1 Nonmetals. For external parts of sealed components and all parts of unsealed components, insulating and sealing compound shall conform to MIL-S-8660.

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3.2.2.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.2.3 Solder. Solder shall conform to composition Sn50 or Sn60 of QQ-S-571. A rosin flux shall be used.

3.3 Design and construction. The relay assembly shall conform to this specification and to the drawing number 8338636 listed under army drawings in 2.1.2 (see 4.8.2).

3.3.1 Soldering. Soldering shall meet the requirements of DOD-STD-1866.

3.3.2 Standard parts. Military standard parts shall be incorporated wherever applicable. Commercial standard parts may be used, provided they are interchangeable with Military Standard parts without modification, and the equivalent MS part is identified by the manufacturer.

3.3.3 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.3.4 Springs. Springs shall not be used to transmit current.

3.3.5 Threaded parts. All screw threads shall conform to FED-STD-H28.

3.3.6 Polarity. The relay assembly shall operate in a circuit having the negative battery terminal grounded.

3.3.7 Grounding. The finish around the mounting holes shall provide positive electrical contact to ground when the assembly is normally mounted. This requirement does not apply to relay assemblies mounted on insulated bases.

3.3.8 Mounting bracket. When brackets are used, they shall be an integral part of the housing or shall be securely attached thereto, in a manner to prevent any movement between the assembly and the mounting bracket in service.

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3.3.9 Weight. The maximum weight of the relay assembly shall be four pounds.

3.4 Performance.

3.4.1 Time delay. The relay windings and related parts shall provide a time delay (time the main gun-firing relay contacts are closed) between 0.10 and 0.50 seconds under all conditions of operations throughout the operating voltage range of 18 to 30 V dc in each of its three major axes. The relay assembly shall subsequently meet the requirements specified in 3.4.2 through 3.4.9 (see 4.8.3).

3.4.2 Extreme temperature operation. The relay assembly shall meet the requirement 3.4.1 when tested at air temperatures of minus 65 ± 5 degrees Fahrenheit (°F) and plus 160 ± 5°F (see 4.8.4).

3.4.3 Operating current. When the main gun relay is energized, the operating current shall not be more than 0.35 ampere (amp) (see 4.8.5).

3.4.4 Dielectric strength. When tested as specified in 4.8.6, the relay assembly shall evidence no loosening, cracking, burning, smoking, rupturing or any other damage to the insulation. It shall subsequently meet the requirements of 3.4.1, 3.4.6, 3.4.7, and 3.4.9 (see 4.8.6).

3.4.5 Endurance. When tested as specified in 4.8.7, the relay assembly shall remain mechanically and electrically operative and the voltage drop across the main gun and machine gun relay contacts carrying rated current, shall be as specified in 4.8.11 and shall not exceed the limits specified in 3.4.9. The relay assembly shall subsequently meet the requirements of 3.4.1, 3.4.6, 3.4.7 and 3.4.9 (see 4.8.7).

3.4.6 Pickup voltage. The pickup voltage of the machine gun relay shall not exceed 18 V dc when tested for ten cycles (deenergize and energize) (see 4.8.8).

3.4.7 Drop-out voltage. The drop-out voltage of the machine gun relay shall be between 1.5 to 8 V dc (see 4.8.9).

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3.4.8 Coil Current. The coil current of the machine gun relay shall be not more than 0.35 amp (see 4.8.10).

3.4.9 Contact voltage drop. Unless otherwise specified on the applicable military standard or drawing, the voltage drop across external terminals of the main gun and machine gun relay assemblies, shall not exceed individual reading of 175 millivolts or an average of 150 millivolts before cycling and individual reading of not more than 200 millivolts or an average of 175 millivolts after cycling tests (see 4.8.11).

3.4.10 Environmental conditions. The design, materials, and construction of relay assembly shall enable the unit to operate in the following environmental conditions. The relay assembly shall subsequently meet the requirements specified in 3.4.1, 3.4.2, 3.4.6, 3.4.7, and 3.4.9 (see 4.8.12).

3.4.10.1 Vibration. The relay assembly shall evidence no mechanical or electrical malfunction (opening or closing of relay contacts) when tested in accordance with method 201 of MIL-STD-202 (see 4.8.12.1).

3.4.10.2 Shock. The relay assembly shall evidence no mechanical or electrical malfunction (opening or closing of relay contacts) when tested in accordance with method 213 of MIL-STD-202. (see 4.8.12.2).

3.4.10.3 Corrosion. The relay assembly shall evidence no corrosion during or after salt spray treatment when tested in accordance with method 101 of MIL-STD-202, except that the test duration shall be 200 hours (see 4.8.12.3).

3.4.10.4 Waterproofness. The relay assembly shall be waterproof when tested in accordance with type II , class I of MIL-STD-1184 (see 4.8.12.4).

3.5 Finish. (see 4.8.2).

3.5.1 Painting. Cleaning, conditioning, and painting shall conform to the requirements of MIL-STD-193.

3.5.2 Electroplating. Exposed, unprotected metal surfaces shall be electroplated in accordance with Class 2, type II of QQ-P-416.

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3.5.3 Color. Exterior surfaces of the relay assembly housing shall painted with a white gloss enamel. Color shall conform to color 17875 of FED-STD-595.

3.6 Marking.

3.6.1 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as a minimum shall include the following (see 4.8.2):

Relay Assembly: Gun-Firing
24 V dc Waterproof
Military Part Number (see 6.2)
Manufacturer's Name
Manufacturer's Serial Number
or Date of Manufacture (Month, Year, e.g.,
October 86)
US

3.7 Workmanship. Workmanship shall be such as to assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective receptacle, cracked insulation, or faulty soldering (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality

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conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662. Required measuring instrument tolerances are as listed in Table I.

4.1.2.1 Apparatus. In addition to standard measuring equipment (see Table I and applicable test specifications), test apparatus shall include a testing chamber capable of maintaining specified air temperature over the range from minus 60 to plus 165 °F. Air movement in the temperature chamber shall be held to a minimum to prevent a temperature gradient within the chamber and any cooling resulting from excess air motion, also a submersion tank capable of producing and maintaining a pressure of 6 pounds per square inch (psi) and a vacuum of 6 psi.

TABLE I. Measuring instrument tolerance.

Measuring Instrument	Accuracy
Voltmeter dc	0.25% (40 volt full scale)
Millivoltmeter dc	0.50% (1000 millivolts full scale)
Ammeter dc	0.50% (0.5 amp full scale) (30 amp full scale) (50 amp full scale) (ability to reset)
Occilloscope	3.0%
Thermometer	1.0%

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4.2 Classification of inspection:

a. First Article Test (see 4.4).

b. Quality conformance inspections (see 4.5).

1. Examination (see 4.5.2).

2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 ± 15 °F

b. Barometric pressure 28.5 + 2.0 - 3.0 inches mercury (Hg)

c. Relative humidity 50 ± 30%

4.4 First article inspection: Unless otherwise specified (see 6.2), five relay assemblies shall be utilized for first article test (see Table II). The relay assemblies shall be representative of the units proposed to be furnished under the contract. First article testing shall be conducted under Government surveillance by the contractor, or an authorized testing facility, at a site approved the Government. Test shall be performed in the order listed in Table II. One sample is to be held in reserve (see 3.1 and 6.2). Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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TABLE II. Order of first article test

TITLE	REQUIREMENT	TEST
SAMPLE 1		
MATERIAL	3.2	4.8.1
DEFECTS (SEE 4.8.2 AND TABLE III)	3.3, 3.5, 3.6 AND 3.7	4.8.2
EXTREME TEMPERATURE OPERATION	3.4.2	4.8.4
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
VIBRATION	3.4.10.1	4.8.12.1
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
SHOCK	3.4.10.2	4.8.12.2
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
SAMPLE 2		
MATERIAL	3.2	4.8.1
DEFECTS (SEE 4.8.2 AND TABLE III)	3.3, 3.5, 3.6 AND 3.7	4.8.2
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
VIBRATION	3.4.10.1	4.8.12.1
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
WATERPROOFNESS	3.4.10.4	4.8.12.4
DIELECTRIC STRENGTH	3.4.4	4.8.6
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11

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TABLE II. Order of first article test-Continued.

TITLE	REQUIREMENT	TEST
SAMPLE 3		
MATERIAL	3.2	4.8.1
DEFECTS (SEE 4.8.2 AND TABLE III)	3.3., 3.5, 3.6 AND 3.7	4.8.2
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
OPERATING CURRENT	3.4.3	4.8.5
COIL CURRENT	3.4.8	4.8.10
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGER	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
SAMPLE 4		
MATERIAL	3.2	4.8.1
DEFECTS (SEE 4.8.2 AND TABLE III)	3.3, 3.5, 3.6 AND 3.7	4.8.2
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11
ENDURANCE TEST	3.4.5	4.8.7
TIME DELAY	3.4.1	4.8.3
PICKUP VOLTAGE	3.4.6	4.8.8
DROP-OUT VOLTAGE	3.4.7	4.8.9
CONTACT VOLTAGE DROP	3.4.9	4.8.11

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4.4.1 Failure. Failure of a first article sample to pass any of the inspections specified herein may be cause for the Government to refuse to conduct additional inspections until the faults revealed by the inspection have been correct.

4.5 Quality Control Inspection (QCI). QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all (name of item) of a single type, class, style and part identification number (PIN), from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table III.

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TABLE III. Sampling plan for QCI.

QCI sampling plan			
Inspection lot size	Sample size		
	Examination 1/		Test/2
	Major	Minor	
2 to 8	*	5	
9 to 15	13	5	
16 to 25	13	5	
26 to 50	13	5	
51 to 90	13	7	
91 to 150	13	11	
151 to 280	20	13	
281 to 500	29	16	
501 to 1200	34	19	
1201 to 3200	42	23	
3,201 to 10,000	50	29	
10,001 to 35,000	60	35	
35,001 to 150,000	74	40	
150,001 to 500,000	90	40	
500,001 to over	102	40	

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in Table IV (see 4.7.2). The acceptance number in all cases is zero.

4.5.2.1 Classification of defects. For examination purposes, defects shall be classified as listed in Table IV.

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TABLE IV. Classification of defects.

Category	Defect	Method of Examination
Critical	None	
Major		
101	Dimensions out of tolerance (see 3.3)	SIE 1/
102	Marking, improper (see 3.6)	Visual
Minor		
201	Dimensions, out of tolerance (see 3.3)	SIE 1/
202	Finish, improper (see 3.5)	Visual
203	Workmanship, faulty (see 3.7)	Visual

1/ SIE - Standard Inspection Equipment

4.5.3 Tests. Samples selected in accordance with 4.5.1.2 shall be subjected to the Quality Conformance Tests in the test sequence specified in Table V and Table VI.

TABLE V. Order of quality conformance test.

TITLE	REQUIREMENT	TEST
Time Delay	3.4.1	4.8.3
Pick-Up Voltage	3.4.6	4.8.8
Drop-Out Voltage	3.4.7	4.8.9
Contact Voltage Drop	3.4.9	4.8.11
Operating Current	3.4.3	4.8.5
Coil Current	3.4.8	4.8.10
Dielectric Strength	3.4.4	4.8.6
Time Delay	3.4.1	4.8.3
Contact Voltage Drop	3.4.9	4.8.11

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TABLE VI. Order of control test.
(To be performed at ambient temperature of 77 ± 15 °F)

TITLE	REQUIREMENT	TEST
Time Delay	3.4.1	4.8.3
Pick-Up Voltage	3.4.6	4.8.8
Drop-Out Voltage	3.4.7	4.8.9
Contact Voltage Drop	3.4.9	4.8.11
Waterproof Test	3.4.10.4	4.8.12.4
Dielectric Strength	3.4.4	4.8.6
Time Delay	3.4.1	4.8.3
Contact Voltage Drop	3.4.9	4.8.11

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100% inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.6 Control tests.

4.6.2 Sampling for control test. Relay assemblies shall be selected at the rate of three of each 500 produced, or fraction thereof, provided that not more than six units shall be selected in any 30-day period. The control test sample shall be identified as to production period, and after having been examined for the defects specified in 4.5.2.1, they shall then be subjected to the tests specified in Table VII in the order listed.

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TABLE VII. Control test sequence.

SAMPLE	TEST SEQUENCE PARAGRAPH NUMBER	TEST DESCRIPTION
1	4.8.7	Endurance
2 and 3	4.8.12.1	Vibration
	4.8.12.4	Waterproofness
	4.8.12.2	Shock
	4.8.12.3	Corrosion

4.7 Failure. Failure of any relay assembly to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials design and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawing, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.2, 3.3, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in Table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Time delay. To determine conformance to 3.4.1, the relay assembly shall be subjected to test potentials of 18, 24, and 30 V dc. The time delay shall be measured with an oscilloscope or other time interval measuring equipment, accurate to within three percent. These tests shall be repeated with the relay assembly mounted in the direction of each of its three major axes.

4.8.4 Extreme temperature operation. To determine conformance to 3.4.2, the test specified in 4.8.3 shall be repeated at temperatures of minus 65 $\pm$ 5 $^{\circ}$F and plus

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160 ± 5°F. Prior to testing at each temperature, the relay assemblies shall be subjected to that temperature for 24 hours.

4.8.5 Operating current. To determine conformance to 3.4.3, following procedure shall be followed:

- Connect 29 + .00, -.5 V dc between pins B and G of the relay assembly.

- Energize main gun relay by applying 29 V dc between pins C and G of the relay assembly.

- Measure the coil current.

4.8.6 Dielectric strength. To determine conformance to 3.4.4, relay assembly shall be subjected to the test specified in method 301 of MIL-STD-202. Applied voltage shall be of 500 volts root mean square (RMS) at a frequency of 60 Hertz. The test voltage source shall have a minimum rating of 0.5 Kilovolt-Ampere. Voltage shall be applied as follows:

a. Between the winding terminals and the housing. If necessary winding internal grounds could be disconnected.

b. Between the terminals of the relay assembly that are connected to the contact points, and the relay housing.

The test potential shall be applied and reduced at a maximum rate of change of 250 volts per second. Duration of application of the test voltage shall be one minute for each winding and for the circuit breaking, mechanisms. During which time the relay assembly shall be observed for evidence of short circuit or insulation break down. The relay assembly shall subsequently be subjected to the tests specified in 4.8.3.

NOTE: Dielectric strength test could be conducted prior to final assembly of relay assembly 8338636.

NOTE: Before high voltage is impressed upon any winding, terminal or other part of the relay assembly, the construction should be carefully examined and the manufacturer's internal circuit diagram for the device checked to determine where windings and other parts are grounded. The appropriate low voltage circuit ground and

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circuit continuity test should be made to assure that high voltage may not, through some unintended circuit, cause serious damage, instead of determining insulation effectiveness.

4.8.7 Endurance test.

4.8.7.1 Main gun relay. To determine conformance to 3.4.5, the main gun relay assembly shall be subjected to 50,000 cycles of operation at 24 V dc. Cycling shall consist of making and breaking a 45 amp dc load in an inductive circuit with the time constant ratio L/R (time required for the current to reach 63.2% of its steady state value) not less than .026 second. On time shall be 0.3 ± .2 second. Off time shall be 5.7 ± 0.2 seconds. Voltage drop across the contact points shall be measured after the test with a millivoltmeter accurate within 0.5%. The 50,000 cycles of operation shall be conducted at the following temperatures:

NUMBER OF CYCLES	TEMPERATURES
20,000	77 ± 15 °F
15,000	-65 ± 5 °F
15,000	160 ± 5 °F

The relay assembly shall then be subjected to the test specified in 4.8.3 and 4.8.11.

4.8.7.2 Machine gun relay. To determine conformance to 3.4.5, the machine gun relay shall be subjected to the test specified in 4.8.7.1, except that the following values shall be substituted. Cycling shall consist of making and breaking a 20 amp dc inductive load. On time shall be 4 ± 0.1 seconds. Off time shall be 8 ± 0.1 seconds. The machine gun relay shall then be subjected to the test specified in 4.8.10 and 4.8.11.

4.8.8 Pick-up voltage. To determine conformance to 3.4.6, after the machine gun relay has been temperature stabilized for 24 hours, it shall then be tested for 10 cycles (deenergize and energize) to determine that the energized function is completed when a potential not in excess of the pick-up voltage as specified in 3.4.6 is applied. A suitable indicating device shall be used to determine that the contacts operate satisfactorily. Upon completion of this test, the test shall be repeated at ambient air temperatures of minus 65 ± 5°F and plus 160 ± 5°F.

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4.8.9 Drop-out voltage. To determine conformance to 3.4.7, after the machine gun relay has been temperature stabilized, it shall then be energized with 28 V dc. The voltage shall then be reduced. The relay shall release to the deenergized position within the specified limits of drop-out voltage as specified in 3.4.7. Repeat for 10 cycles. A suitable device shall be used to determine that the contacts operate satisfactorily. Upon completion of this test, the test shall be repeated at temperature of minus 65 ± 5°F, and plus 160 ± 5°F.

4.8.10 Coil current. To determine conformance to 3.4.8, the machine gun relay shall be temperature stablized. A voltage of 29 + 00, -.50 V dc shall be applied between pins E and G of the relay assembly, and the current flow determined within 10 seconds after application of coil voltage.

4.8.11 Contact voltage drop. To determine conformance to 3.4.9, an average of 10 consecutive and individual readings shall be within the limits as specified in 3.4.9. Contacts shall be caused to break and then to make the test current before each measurement. The machine gun relay shall be subjected to a 20 amp dc resistive load across the contacts and 28 V dc on the coil. The main gun relay shall be subjected to a 45 amp dc resistive load across the contacts and 28 V dc on the coil. The voltage drop shall be measured within .05 seconds after contacts close.

4.8.12 Environmental conditions. After each of the following environmental tests, the relay assembly shall be subjected to the tests specified in 4.8.3, 4.8.8, 4.8.9, and 4.8.11, except for the control tests which shall be conducted only at a temperature of 77 ± 15°F.

4.8.12.1 Vibration. To determine conformance to 3.4.10.1, the relay assembly shall be rigidly mounted (without brackets or shock absorbers) and subjected to the vibration test specified in method 201 of MIL-STD-202, except that vibration shall be for one hour along each of the three major axes. Pins E and G of the machine gun relay coil shall be connected to 28.5 V dc power source for half of the test period. To determine electrical malfunction of the relay assembly contact opening or closing shall be monitored between pins A and D of relay assembly.

4.8.12.2 Shock. To determine conformance to 3.4.10.2, the relay assembly shall be rigidly mounted (without brackets or shock absorbers) and subjected to the shock test specified

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in method 202 of MIL-STD-202 except that the shock impulses shall be 25 ± 3 gravity units, 6 to 9 milliseconds duration, half sine wave, applied in each direction of three mutually perpendicular axes. When the relay assembly is tested in energized position the maximum contact opening duration shall not exceed two milliseconds. This test shall be conducted once in energized position and once in deenergized position of the relay assembly.

4.8.12.3 Corrosion. To determine conformance to 3.4.10.3, the relay assembly shall be subjected to the salt spray (corrosion) test specified in method 101 of MIL-STD-202, except that the test duration shall be 200 hours. a 5.0% salt solution shall be used.

4.8.12.4 Waterproofness. To determine conformance to 3.4.10.4, the relay assembly shall be subjected to the test for waterproofness in accordance with type II, class 2 of MIL-STD-1184.

5. PACKAGING.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gun-firing relay assembly covered by this specification is intended for the protection of the gun-firing solenoid on Military combat vehicles from burn-out, by preventing the solenoid from remaining continuously energized.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Item, name and part number (see 3.3).

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c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

d. If responsibility for inspection shall be other than as specified (see 4.1).

e. If responsibility for inspection equipment shall be other than as specified (see 4.3).

f. If first article test samples are other than as specified (see 4.4).

g. Selection of applicable level and packaging requirements (see 5).

6.3 First Article. When first article test is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4 and 4.4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in

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hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot or inspection batch, i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use of operation of the unit

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

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l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample. A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.3).

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows, or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et

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seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42U.S.C 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Subject Term (key word) listing.

- Gun firing relay assembly, waterproof, 24 V dc.
- Waterproof relay assembly, 24 V dc, gun firing.

6.6 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

Custodian:
Army - AT

Preparing activity:
Army - AT

User activity:
Army - AR

(Project 5945-A726)

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